Well Posed Learning Problem – A computer program is said to learn from experience E in context to some task T and some performance measure P, if its performance on T, as was measured by P, upgrades with experience E.

Any problem can be segregated as well-posed learning problem if it has three traits –

- Task
- Performance Measure
- Experience

Certain examples that efficiently defines the well-posed **learning problem are –**

1. To better filter emails as spam or not

- Task – Classifying emails as spam or not
- Performance Measure – The fraction of emails accurately classified as spam or not spam
- Experience – Observing you label emails as spam or not spam

2. A checkers learning problem

- Task – Playing checkers game
- Performance Measure – percent of games won against opposer
- Experience **–** playing implementation games against itself

3. Handwriting Recognition Problem

- Task – Acknowledging handwritten words within portrayal
- Performance Measure – percent of words accurately classified
- Experience – a directory of handwritten words with given classifications

4. A Robot Driving Problem

- Task – driving on public four-lane highways using sight scanners
- Performance Measure – average distance progressed before a fallacy
- Experience – order of images and steering instructions noted down while observing a human driver

5. Fruit Prediction Problem

- Task – forecasting different fruits for recognition
- Performance Measure – able to predict maximum variety of fruits
- Experience – training machine with the largest datasets of fruits images

6. Face Recognition Problem

- Task – predicting different types of faces
- Performance Measure – able to predict maximum types of faces
- Experience – training machine with maximum amount of datasets of different face images

7. Automatic Translation of documents

- Task – translating one type of language used in a document to other language
- Performance Measure – able to convert one language to other efficiently
- Experience – training machine with a large dataset of different types of languages